BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 099 · Storage Units and Conversions

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What should guide the choice of a storage device?

Q2. When does a compiler translate source code?
A. When a key is pressed
B. Only after the program ends
C. Before the program runs
D. Only while each line runs

Q3. For scores = [5, 1, 3], total starts at 0 and each element is added once. Give the successive totals and the final result.

Q4. Use decimal units. Convert 5 TB to GB, then state how many TB are in 1 PB.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

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I Do · Read new knowledge

Key vocabulary

Bit — A binary digit, either 0 or 1.

Nibble — Four bits.

Byte — Eight bits.

Kilobyte — 1000 bytes using decimal SI units.

Megabyte — 1000 kilobytes using decimal SI units.

New learning

Small units: A bit stores one binary digit; a nibble is four bits and a byte is eight bits.

Larger units: For GCSE storage-capacity questions here: 1 kB = 1000 bytes; 1 MB = 1000 kB; 1 GB = 1000 MB. State assumptions where relevant.

Conversion direction: Moving to a smaller unit multiplies; moving to a larger unit divides. Bits and bytes require a factor of eight.

Capacity calculations: Convert all sizes to the same unit before dividing total capacity by file size.

Worked example

Convert 4 MB into kB and then into bytes using decimal units. [4 marks]

1. State the conversion factor

2. 1 MB = 1000 kB and 1 kB = 1000 bytes.

3. 4 × 1000 = 4000 kB.

4. 4000 × 1000 = 4,000,000 bytes. [4]

We Do · Practise with guidance

Guided 1

Convert 3 GB into MB and kB. Show both steps. [4]

Worked method: 3 × 1000 = 3000 MB [2]; 3000 × 1000 = 3,000,000 kB [2].

Guided 2

Convert 56 bits to bytes and explain the operation. [3]

Worked method: There are 8 bits in a byte [1]; 56 ÷ 8 [1] = 7 bytes [1].

Guided 3

A 5 GB drive stores files of 250 MB each. Calculate how many fit. [4]

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The application's requirements, including capacity, speed, cost and physical use.

Q2. C — Before the program runs

Q3. 5, 6, 9; final total 9.

Q4. (a) 5,000 GB; 1 PB = 1,000 TB.

I Do / We Do pages · Guided practice

Q1. 3 × 1000 = 3000 MB [2]; 3000 × 1000 = 3,000,000 kB [2].

Q2. There are 8 bits in a byte [1]; 56 ÷ 8 [1] = 7 bytes [1].

Q3. 5 GB = 5000 MB [1]; 5000 ÷ 250 [1] = 20 files [1], assuming no overhead [1].

You Do pages · Code First

```
bits = 43
leftover = bits MOD 8
print(leftover)
```

Expected result: 3

You Do pages · Independent questions

Q1. 2.5 × 1000 [1] = 2500 kB [1].

Q2. 12 × 8 [1] = 96 bits [1].

Q3. 6 GB = 6000 MB [1]; 6000 ÷ 750 [1] = 8 whole files [1], ignoring overhead.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: 5 GB = 5000 MB [1]; 5000 ÷ 250 [1] = 20 files [1], assuming no overhead [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Display the number of leftover bits when 43 bits are grouped into bytes.

```
bits = 43
leftover = ...
print(leftover)
```

write

Question 1

Convert 2.5 MB into kB using decimal units. [2] Source: Original OCR J277-style practice question.

Question 2

Calculate the number of bits in 12 bytes. [2] Source: Original OCR J277-style practice question.

Question 3

A device has 6 GB free. Each file uses 750 MB. Calculate the maximum whole files it can store. [3] Source: Original OCR J277-style practice question.

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

An algorithm only gives reliable results when its inputs are handled as specified.